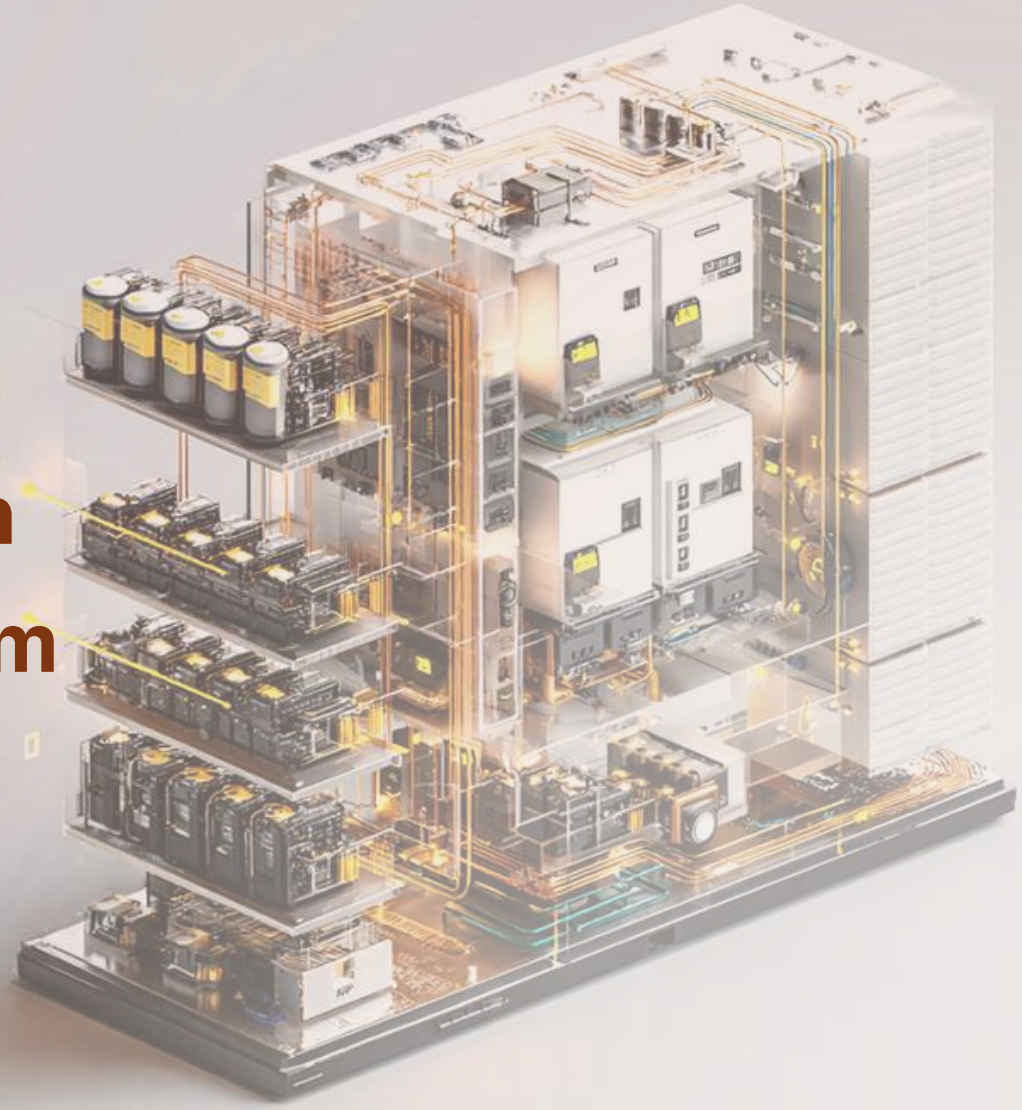
PIOST

Advancing Sustainable Thermal Energy Storage Technology for Carbon-Neutral Heating Solutions

Delivering Next-Generation Hot Water and Heating through Thermal Energy Storage System



PIOST Inc. | Junbeom Song (wnsqja0718@inha.edu)

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Overview

PIOST is replacing fossil-fuel boilers with a clean, renewable-energy based heating

Item Overview

- Heating and hot-water supply system based on thermal energy storage
- Long-duration heat supply by storing renewable electricity as heat energy

Core Technologies

Rock & sand heat storage medium

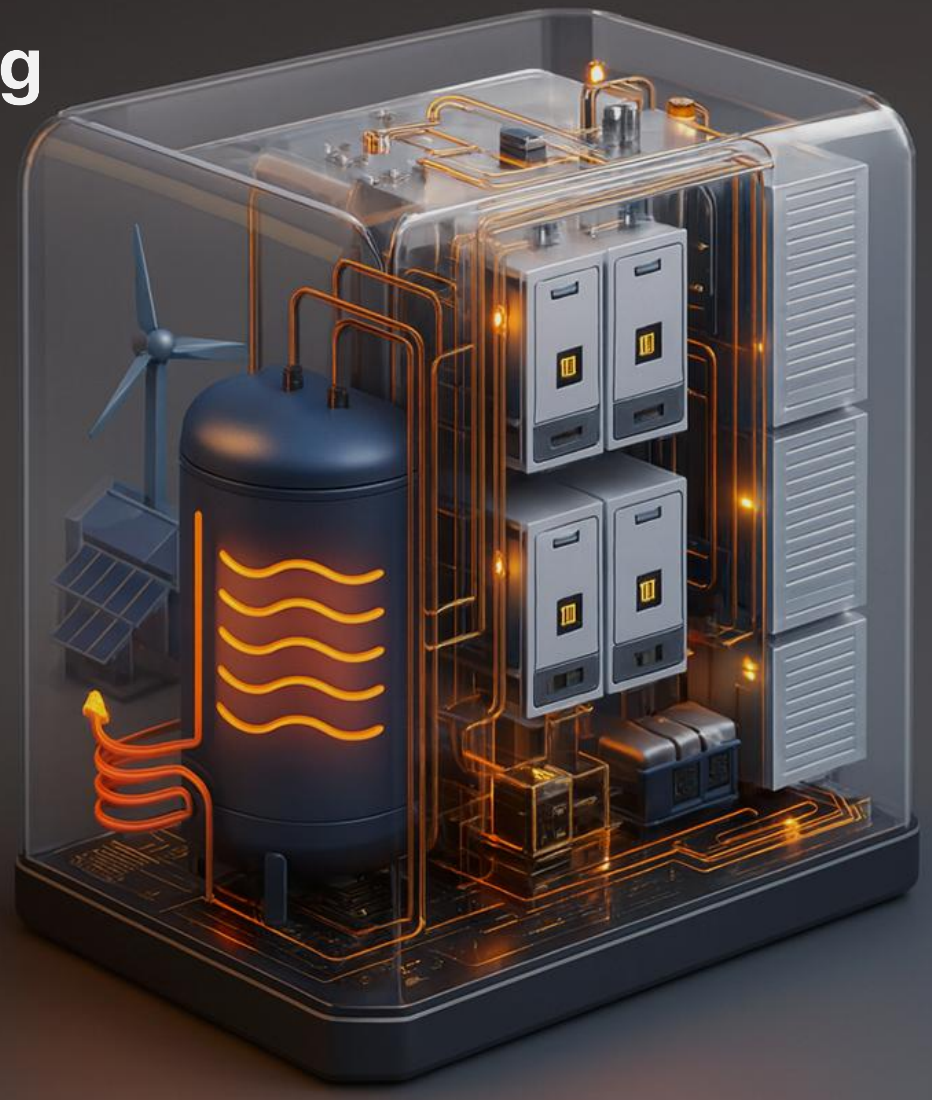
Direct condensation heat-exchange

Modular design for scalability

High-temperature energy storage

Market Outlook

- The global TES market is projected to grow more than tenfold by 2035 (McKinsey)
- In Europe, PV capacity is projected to grow to over 670GW by 2028 (EEA)
- Renewables accounted for about 25.4% of the EU' final energy consumption (SPE)

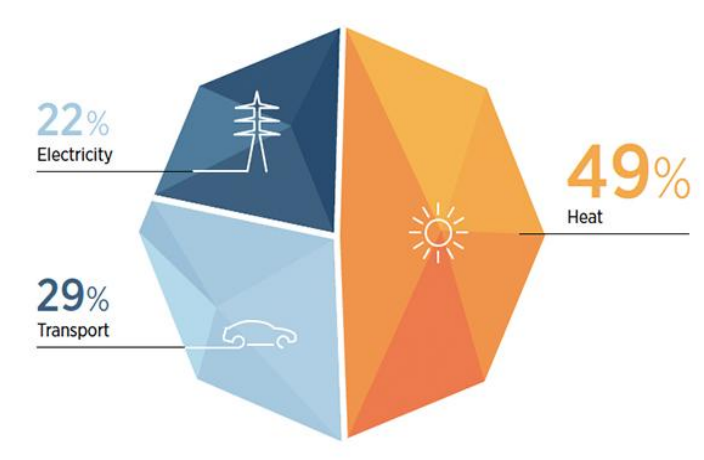


Problem

Thermal energy storage is essential for renewable-based carbon reduction

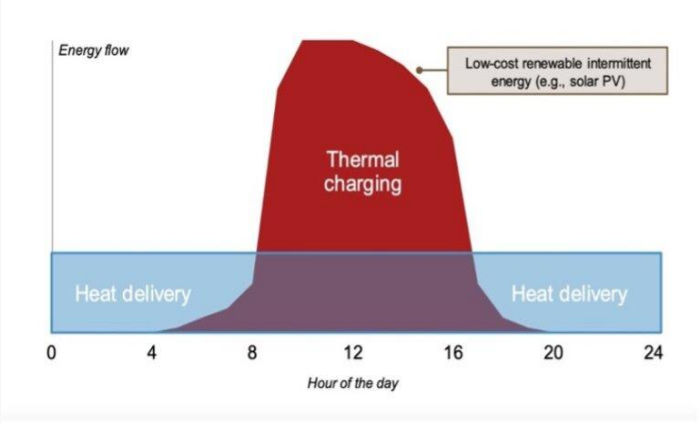
TES, a solution to renewable intermittency and carbon emissions from fossil-fuel based heating

Heating demand with energy transition



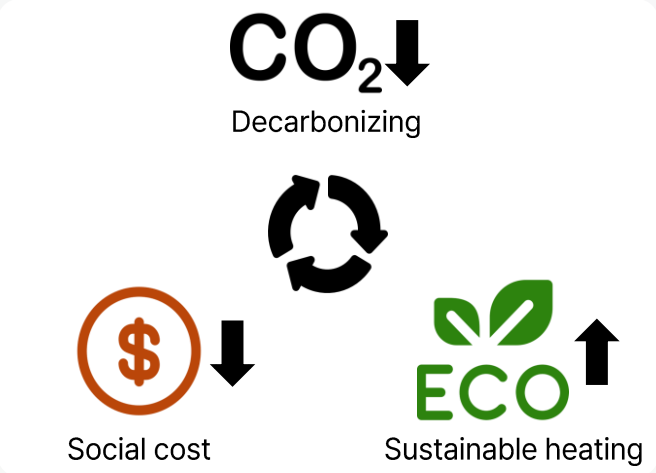
- Heat accounts for half of global energy demand
- Decarbonizing heating is key to completing the energy transition

Need for long-duration energy storage



- Ensuring stable heat supply despite renewable fluctuation
- Storing heat by day, delivering when needed

Social and industrial impact



- Enabling sustainable, eco-friendly heating
- Supports EU ETS and Fit for 55 goals by reducing CO_2 and energy costs

Solution

We are demonstrating TES abroad to enable European market

Conducting overseas testbed demonstrations to validate TES performance and prepare for European deployment

- Stepwise roadmap for verification, localization and commercialization
- Phase 1 (2025): Complete TES prototype and conduct pilot demonstration under cold-climate conditions (e.g., Northern Europe)
- Phase 2 (2026): Collaborate with regional partners to adapt system design to European standards
- Phase 3 (2027): Expand deployment to European B2B/B2G markets, focusing on public facilities and industrial heat users

<IR pitching for Mongolia pilot collaboration>

<MOU signing for Mongolia testbed partnership>

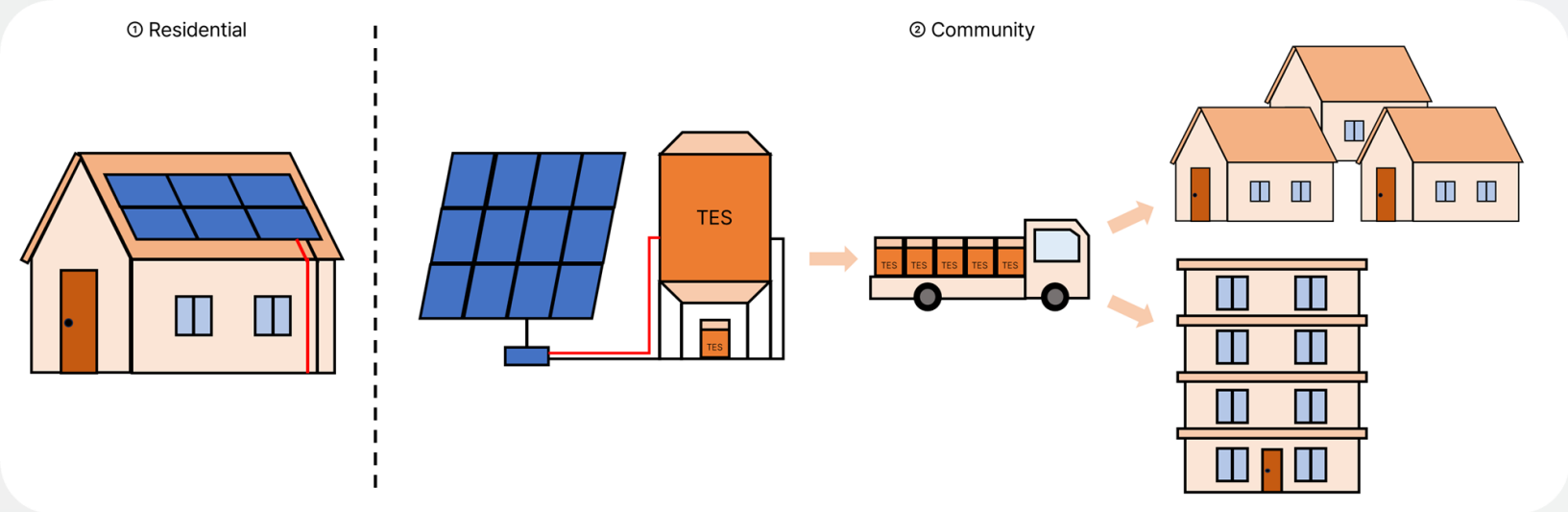
<Business meeting for TES adaptation>

Solution

We are customizing TES heating models: from single unit home to industry

Expanding modular TES systems adaptable to diverse European environments

- TES applications by customer type
- Type 1 (Residential): Off-grid heating solution for individual homes
- Type 2 (Community): Integrated TES systems for residential clusters and shared heating networks



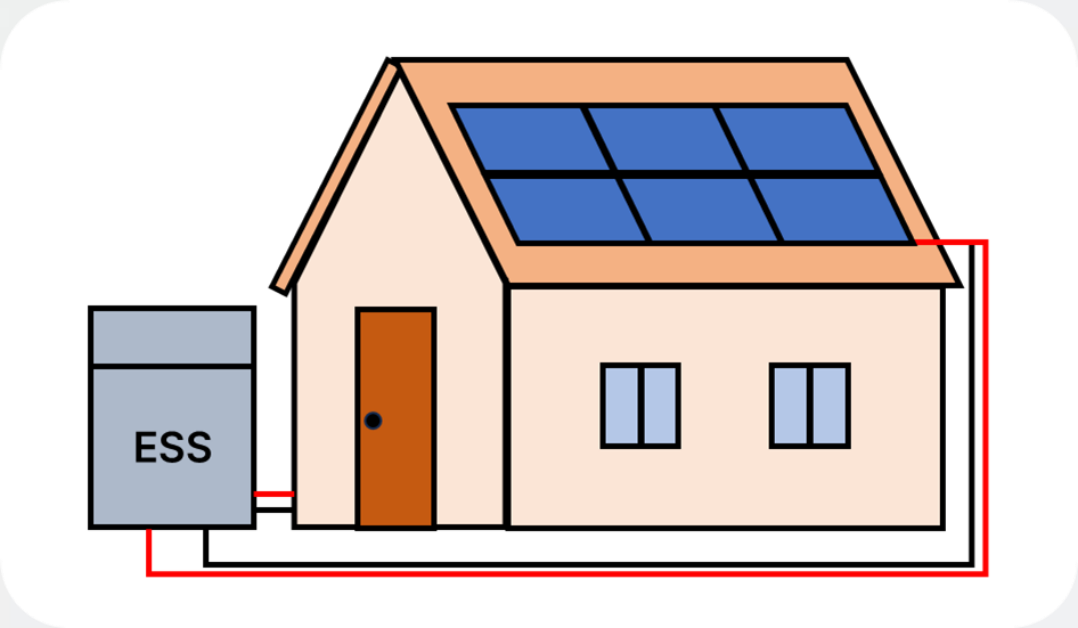
<Expansion of TES-based heating models from residential to community applications>

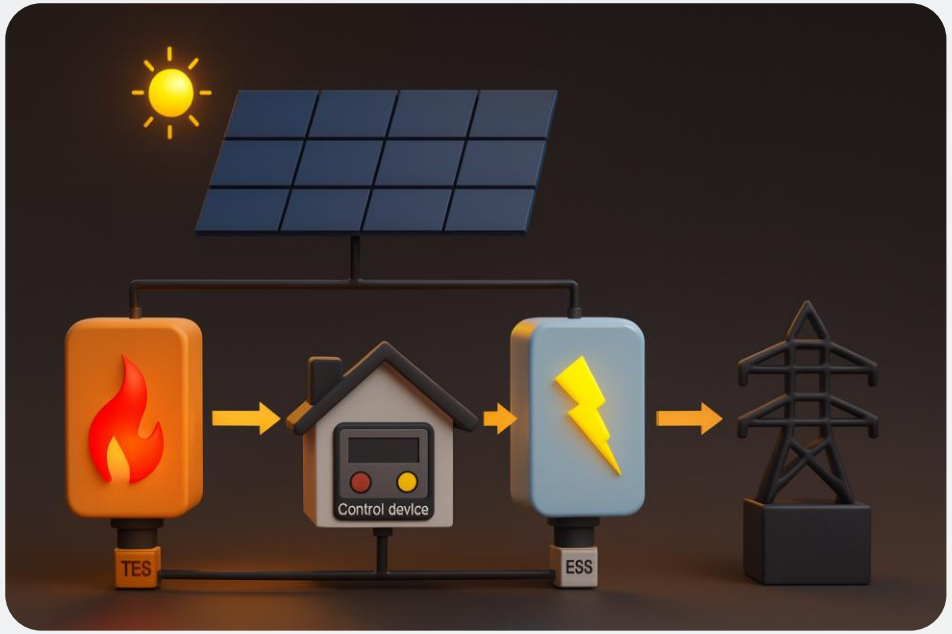
Solution

We are suggesting an integrated ESS-TES energy solution

Building an integrated heating and power storage solution for sustainable communities

- ESS (Electricity Storage System) and TES (Thermal Energy Storage) are combined to form a unified energy hub
- The integrated system enables simultaneous storage and supply of heat and electricity, improving energy utilization efficiency
- Operates using renewable energy (PV, wind) to achieve low-carbon, grid-independent operation





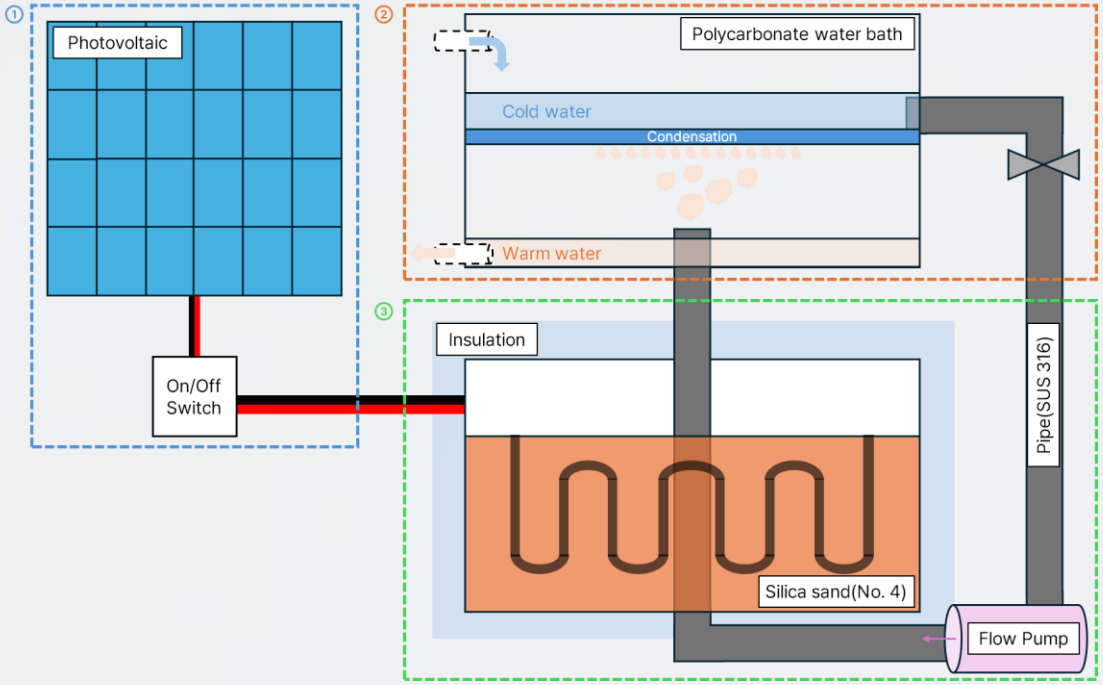
<Illustration: TES-ESS Integrated Hybrid Energy Network>

Solution

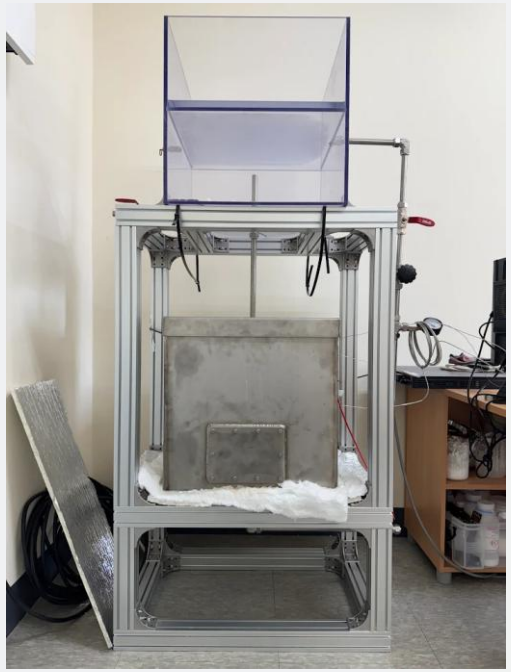
We are making innovations securing global market readiness

Advancing experimental TES technologies and strengthening intellectual property (IP) portfolio

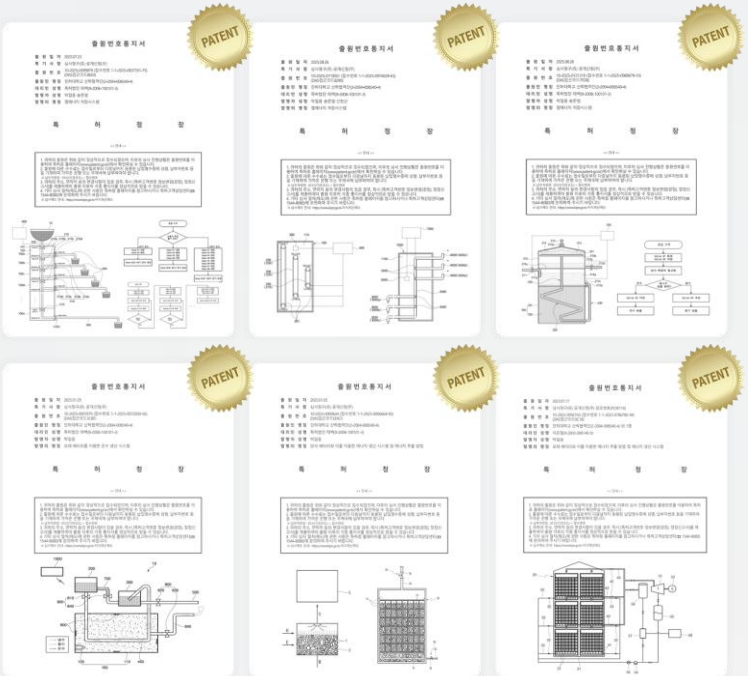
- Technology-driven competitiveness strategy
  - Enhance four core TES technologies: Rock & sand medium, Direct condensation heat-exchange, Modular design and High-temperature system
  - Expand domestic and international patent portfolios through PCT filings to strengthen IP protection and facilitate overseas market entry



<Conceptual diagram of TES system operation>



<Lab-scale prototype and test setup>



<Patent portfolio for technology protection>

Scale-up

We are securing sustainable growth by stepwise funding and investment

Roadmap from R&D support to private investment and market expansion

2025	2026 – 2027	2027 – 2028	2028 – 2029
Inha Univ. joint R&D	Attract seed/accelerator funding	Secure venture capital investment	Explore ODA and climate-tech funding opportunities
Mongolia B2C revenue model	Aim to Horizontal Europe, EIC Accelerator	Expand B2B/B2G cooperation	Generate revenue through European B2B/B2G model

Initial market entry

- Validate TES performance through European pilot operations
- Establish early B2C partnerships with local companies
- Obtain product certification and export approval based on pilot data

Revenue expansion

- Commercialize modular TES systems and O&M service models
- Introduce Heating-as-a-Service (HaaS) subscription business model
- Expand B2G contracts with European public institutions

Long-term growth

- Localize production and establish mass-manufacturing lines
- Diversify exports across Central and Northern Europe
- Contribute to regional net-zero energy communities

Scale-up

We are capturing market leadership through superior TES

Performance comparison among domestic and overseas competitors

Category	Domestic competitor	Overseas A	Overseas B	PIOST Inc.
Max temp.	500 [°C]	600 [°C]	650 [°C]	**Upper 800[°C]**
Storage material	Molten salt	Sand	Sand	**Composite rock & sand** for cost advantage
Heat transfer	Indirect	Indirect	Indirect	**Direct condensation & heat exchange** for enhancing heat transfer rate
System type	Unit	Plant	Unit	**Modular system** for scalability and ease of deployment

Team

We are an organization bridging R&D and commercialization

Core Management Team

- A multidisciplinary founding team integrating R&D, market analysis and business strategy

Dr. Junbeom Song
CEO

- Market analysis and business strategy development
- Investment promotion

Prof. Il Woong Park
CTO

- R&D planning
- Resource management

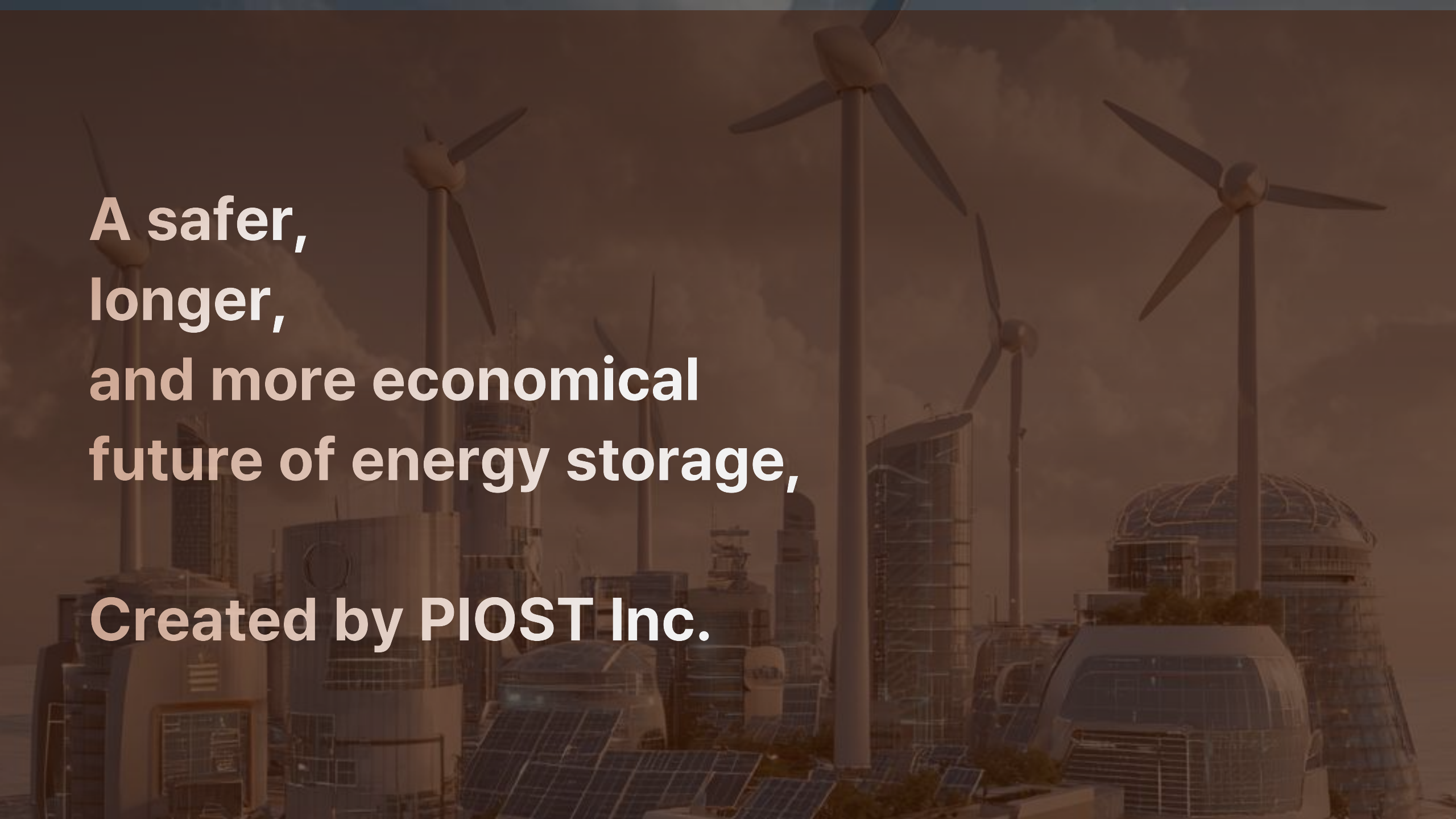
Unjang Lee
Senior advisor

- Providing technical and strategic advisory across commercialization

Research Team

- Experiment-oriented specialists enhancing technology maturity from lab to field application

Hyunjin Yong	Jaehyeok Yang	Sangmin Song	Hyungeun Shin	Sungjin Kim	Kyeongju Ko	Chaeyeon Kim	Eunbin Park



A safer,
longer,
and more economical
future of energy storage,

Created by PIOST Inc.